

Activity #1 – Variables

Instructions

- Carefully analyze the given Java program.
- Trace the references and memory changes step-by-step.
- Complete all required tables and questions.
- Show your logical reasoning where necessary.
- Write your answers in a yellow paper (handwritten).
- Take a clear photo or scan of your work.
- Upload your file in Canvas on or before the deadline.
- Submission Format: Section_LastName_FirstName_CodeAnalysisActivity#1.pdf

Please review the code below.

```
class Student {
    String name;
    int grade;

    Student(String n, int g) {
        name = n;
        grade = g;
    }
}

public class TestStudent {
    public static void main(String[] args) {

        Student s1 = new Student("John", 85);
        Student s2 = new Student("Maria", 90);
        Student s3 = s2;

        s1.grade = 95;
        s3.name = "Ana";

        System.out.println(s1.name + " " + s1.grade);
        System.out.println(s2.name + " " + s2.grade);
        System.out.println(s3.name + " " + s3.grade);
    }
}
```

Please answer the following questions:

PART A – VARIABLE IDENTIFICATION TABLE

Variable Name	Data Type	Variable Type	Stored In

PART B: Explanation. Please see example below.

Statement	What Happens?	Memory Impact
<code>Student s1 = new Student("John", 85);</code>		
<code>Student s2 = new Student("Maria", 90);</code>		
<code>Student s3 = s2;</code>		
<code>s1.grade = 95;</code>		
<code>s3.name = "Ana";</code>		

PART C – OUTPUT